



Republic of Rwanda
Ministry of Education



RTB | RWANDA
TVET BOARD

LSVQC302

SURVEYING PROJECT QUANTITIES AND COST ESTIMATES

Prepare surveying project quantities and cost estimates

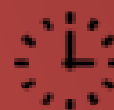
Competence

RQF Level: 3

Learning Hours

Credits: 4

40



Sector: Construction and Building Services

Trade: Land surveying

Module Type: Specific

Curriculum: CBSLSV3001 - TVET Certificate 3 in Land surveying

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Purpose statement	This module is designed for students pursuing level 3 TVET certificate in land surveying. It describes the skills, knowledge, and attitude required to estimate quantity and cost of surveying projects. At the end of this module, the student will be able to estimate quantities, analyze costs and produce an abstract of cost for a surveying project.					
Leaning assumed to be in place	Safety, Health and environment at workplace; Surveying tools and equipment; Surveying communication language; Linear distance measurement; Construction drawings interpretation; Building setting out; Angular measurements.					
Delivery modality	Training delivery		100%	Assessment		Total 100%
	Theoretical content		30%	Formative assessment	30%	50%
	Practical work:		70%		70%	
	• Group project and presentation	20%				
	• Individual project /Work	50%				
			Summative Assessment		50%	

Elements of Competency and Performance Criteria

Elements of competency	Performance criteria	
1. Estimate quantities for surveying projects	1.1.	Items of surveying work are correctly identified with respect to the surveying project.
	1.2.	Items are correctly described with respect to the surveying project
	1.3.	Quantities are accurately calculated based on the surveying project.

2. Analyse costs for surveying projects	2.1. Direct cost is accurately calculated according to the surveying project.
	2.2. Indirect cost is accurately calculated in accordance with the surveying project.
	2.3. Contingency cost is properly analyzed and calculated according to the given surveying project.
3. Produce bills of quantities and costs	3.1. Bill of quantities is correctly elaborated according to the methods of measurement.
	3.2. Unit rates are properly analyzed with respect to the surveying project.
	3.3. Abstract of costs is correctly elaborated based on the method of measurement.

Course content

Learning outcomes	At the end of the module the learner will be able to: 1. Estimate quantities for surveying project 2. Analyse costs for surveying project 3. Elaborate bills of quantities and costs
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Learning outcome 1: Quantities estimation for surveying project	Learning hours: 13
Indicative content	
* Surveying project items (Refer to ISO 230-6) • Materials, tools, instruments, equipment, labour * Description of items • Specification of items • Unit of measurement of each item * Calculation of quantities for items • Rough methods of quantities estimation • Detailed methods of quantities estimation	
Resources required for the learning outcome	

Equipment	Computer, white/blackboard, projector.
Materials	Marker pen, pieces of chalk, notebooks, pieces paper.
Tools	Calculator, pens, markers, pencils
Facilitation techniques	• Group discussion • Group work • Trainer guided • Practical works
Formative assessment methods	• Written assessment • Performing assessment • Product based assessment

Learning outcome 2: Costs analysis surveying project for		Learning hours: 13
Indicative content		
* Calculation of direct costs • Determination of costs of material, labor, equipment, tools, and instrument. * Calculation of indirect costs • Transport, profit, taxes, depreciation, overhead, etc. * Calculation of contingency costs • Types of contingency costs • Analysis and calculation of contingency costs		
Resources required for the indicative content		
Equipment	Computer, white/blackboard, projector.	
Materials	Pieces of chalk, notebooks, pieces of paper.	
Tools	Calculator, pens, pencils, markers.	
Facilitation techniques	• Individual and group work • Practical exercise • Trainer guided • Group discussion	
Formative assessment	• Written assessment • Performing assessment	

methods	• Product based assessment
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Learning outcome 3: Produce bills of quantities and costs		Learning hours: 14
Indicative content		
* Bill of quantities elaboration • Elements of the bill of quantities • Format of the bill of quantities * Unit rates analysis • Types of unit rates • Factors affecting unit rates. • Calculation of unit rates. * Abstract of costs elaboration • Format of abstract of cost • Content of abstract of cost and calculation of total cost		
Resources required for the indicative content		
Equipment	Computer, white/blackboard, projector.	
Materials	Marker pen, pieces of chalk, notebooks, pieces of paper.	
Tools	abstract of cost sheet, calculator, pens, pencils	
Facilitation techniques	• Individual and group work • Trainer guided • Group discussion • Practical works	
Formative assessment methods	• Written assessment • Performing assessment • Product based assessment	

Integrated/Summative assessment (For specific module)

Integrated situation

AMITH TVET School situated in the Kayonza district wants to construct a football playground. The existing terrain is not levelled. The proposed playground must be 100m long and 50m wide. The surveying task to be performed consists of setting the playground boundaries and leveling the terrain. During levelling the interval between pegs must be 10m. The task will be completed using dumpy level, tape measures, hammer, marker pen, strings, and nails.

As a land surveyor, you are hired to conduct surveying-related works.

- ✓ Estimate quantities for all items related to surveying works (Tools, materials, instruments, equipment, and labour).
- ✓ Estimate the cost for each item related to surveying works.
- ✓ Prepare a bill of quantity and cost estimation related to the surveying works to be done.

The maximum time to complete the given task is 3 hours.

Resources

Tools	Calculator, pens, pencils, drawing set
Equipment	Classroom, working site
Materials	Pieces of papers

Assessable outcomes	Assessment criteria (Based on performance criteria)	Indicator	Observation		Marks allocation
			Yes	No	
Learning outcome 1:	1.1. Required items are identified	Required materials are identified.			2
		Required instruments are			2

(34%)		identified.			
		Required equipment is identified.			2
		Required tools are identified.			2
		Required labour is identified.			2
	1.3.Quantities for required items are calculated	The quantity of materials is calculated.			4
		The quantity of instruments is calculated.			4
		The quantity of equipment is calculated.			4
		The quantity of tools is calculated.			3
		The quantity of labour is calculated.			4
Learning outcome 2: (36%)	2.1. Direct cost is calculated	Labor cost is calculated.			3
		Instrument cost is calculated.			3
		Equipment cost is calculated.			3
		Tools cost is calculated.			3
		Labor cost is calculated.			3
	2.2. indirect cost is calculated	Transport cost is calculated.			3
		Profit cost is calculated.			3
		Taxes are calculated.			3
		The depreciation cost is calculated.			3
		Overhead costs are calculated.			3

	2.3 contingency cost is calculated	Contingency cost is analyzed.			3
		Contingency cost is calculated.			3
Learning outcome 3: (30%)	3.1. bill of quantity is elaborated	The format of the bill of quantity is respected.			5
		Sequence elements of the bill of quantity is respected.			5
	3.2. Rates are analyzed	Rates factors are considered.			5
		Appropriate rates are calculated.			5
	3.3 Abstract of costs is elaborated	The format of the abstract of cost is respected.			5
		Abstract of cost is calculated			5
		Time is respected.			5
Total marks		100			
Percentage Weightage		100%			
Minimum Passing line % (Aggregate): 70%					

References:

1. ISO 230-6 Test Code For Machine Tools — Part 6: Determination Of Positioning Accuracy On Body And Face Diagonals (Diagonal Displacement Tests)